

```
#include <iostream>

#include <string>

using namespace std;

int main() {

    int numCourses;

    cout << "=== CGPA CALCULATOR ===\n\n";
    cout << "Enter the total number of courses: ";
    cin >> numCourses;

    double totalPoints = 0.0;
    double totalCredits = 0.0;

    for (int i = 0; i < numCourses; i++) {

        string CourseName;

        string grade;

        double credits;

        double gradePoints = 0.0;

        cout << "\n--- Course " << (i + 1) << " ---\n";

        cin.ignore();

        cout << "Enter Course Name: ";

        getline(cin, CourseName);

        cout << "Enter Credit Hours: ";
```

```
cin >> credits;
```

```
cout << "Enter Grade (A+, A, A-, B+, B, B-, C+, C, C-, D+, D, F): ";
```

```
cin >> grade;
```

```
if (grade == "A+" || grade == "A")
```

```
    gradePoints = 4.0;
```

```
else if (grade == "A-")
```

```
    gradePoints = 3.67;
```

```
else if (grade == "B+")
```

```
    gradePoints = 3.33;
```

```
else if (grade == "B")
```

```
    gradePoints = 3.0;
```

```
else if (grade == "B-")
```

```
    gradePoints = 2.67;
```

```
else if (grade == "C+")
```

```
    gradePoints = 2.33;
```

```
else if (grade == "C")
```

```
    gradePoints = 2.0;
```

```
else if (grade == "C-")
```

```
    gradePoints = 1.67;
```

```
else if (grade == "D+")
```

```
    gradePoints = 1.33;
```

```
else if (grade == "D")
```

```
    gradePoints = 1.0;
```

```
else if (grade == "F")
```

```
    gradePoints = 0.0;
```

```
else {
```

```
        cout << "Invalid Grade!\n";

        continue;
    }

    totalPoints += gradePoints * credits;
    totalCredits += credits;
}

if (totalCredits > 0) {
    double cgpa = totalPoints / totalCredits;

    cout << "\n===== ";
    cout << "\nTotal Credit Hours: " << totalCredits;
    cout << "\nFinal CGPA: " << cgpa;
    cout << "\n===== \n";
}
else {
    cout << "\nNo credit hours entered. CGPA cannot be calculated.\n";
}

return 0;
}
```